Night shift — accelerating matched SHD learning

ar072 · 19 July 2026 · Draft

Digest

The experiment is registered but has not started. The exp069 evidence indicates that adding epochs is too slow for iteration and does not identify the limiting mechanism. This night therefore uses a short, ordered intervention ladder. It promotes the first candidate that improves both cells, freezes that configuration, and tests it for forty epochs. No official SHD test data may be loaded or inspected.

Mandate

Question and fixed comparison

Can one mechanistically motivated, matched change make both Dale-constrained conductance-based (COBA) and pyramidal-interneuron gamma (PING) networks learn SHD materially faster, while preserving a direct comparison between their recurrent cells?

Both cells retain 256 excitatory and 64 inhibitory slots, identical input and readout dimensions, fixed Dale-constrained recurrence, batch size 32, seed 42, the exp069 development split, membrane-mean classification objective, and the registered validation checkpoint rule. COBA disables its inhibitory loop and uses no voltage-gradient dampening. PING enables the loop and uses dampening 1000. Capacity, connectivity, recurrence structure, data ordering, and all settings not named by a candidate remain unchanged and matched.

Baseline trajectory

Cell	Epoch-five accuracy	Epoch-five loss	Best accuracy through forty
COBA	27.94%	2.472	40.2% at epoch 16
PING	32.6%	2.445	44.85% at epoch 38

The final exp069 selected checkpoints remain a required contextual comparison: COBA reached 43.87% and PING reached 48.04% after eighty epochs.

Registered intervention ladder

1. **Temporal resolution.** Change the integration and input-bin width from 1 ms to 2 ms while retaining the 1,000 ms observation window and every other exp069 setting. This halves backpropagation depth while remaining commensurate with the fastest registered synaptic time constant.
2. **Shared input drive.** Only if candidate 1 is killed, restore 1 ms and increase the shared input-weight mean from 0.9 to 1.2. This tests whether weak PING excitatory recruitment limits early learning without tuning one architecture separately.

3. **Optimization.** Only if candidates 1 and 2 are killed, restore the exp069 input settings and increase the matched Adam learning rate from 0.0004 to 0.001. No scheduler, weight decay, loss change, or architecture-specific optimizer is allowed.

Stop the ladder at the first promoted candidate. Do not combine candidates, search intermediate values, or run later candidates after a promotion. Inhibitory balance, gradient flow, and existing time constants remain diagnostics. A new time-constant mechanism or readout would require a separate exact preregistration before implementation or results; this mandate does not authorize an unspecified rescue.

Promotion, final test, and interpretation

Each candidate first passes a local 128-training/128-validation, two-epoch plumbing smoke, then runs both cells for exactly five epochs on the full development-training partition. Promote the first candidate only when, at epoch five, both cells exceed their own exp069 epoch-five validation accuracy by at least three percentage points, neither validation cross-entropy is worse, all losses and gradients are finite, no update is skipped, and the E population in both cells plus the PING I population remain active and below saturation.

Freeze the promoted configuration before extending it. Train both frozen cells for exactly forty epochs from the registered seed and initialisation. Select checkpoints by highest validation accuracy, then lower validation cross-entropy, then earlier epoch. The primary criterion is an improvement of at least three percentage points for **each** cell over that cell's exp069 selected checkpoint through epoch forty. Report the contemporaneous PING-minus-COBA difference as a secondary diagnostic, not as a seeded architecture claim. Also compare both cells with exp069′s eighty-epoch selected checkpoints without selective checkpoint reporting.

Queue and operating contract

Field	**exp070**
Hypothesis	The first registered short-run intervention that clears the promotion gate will improve both COBA and PING selected held-out validation accuracy by at least three percentage points at forty epochs relative to each cell's exp069 trajectory through epoch forty.
Baseline	exp069, seed 42, on the identical 7,340-utterance development-training and 816-utterance validation partition. Comparisons use exp069 at epoch five, its selected checkpoint through epoch forty, and its final eighty-epoch selected checkpoint; the official SHD test is unavailable.
Kill criterion	Kill an attempt for non-finite loss or logits, any skipped update, silence, saturation, partition mismatch, official-test access, or failure of either cell to clear the registered short-run promotion gate. If all three short candidates are killed, stop without a forty-epoch run. Kill the final hypothesis if either promoted cell improves by less than three percentage points over its own exp069 through-epoch-forty selected validation baseline.
Seed count	1

Field	exp070
Budget	One local 128-training/128-validation two-epoch smoke per implemented candidate; at most three matched five-epoch full-data candidate pairs; then at most one matched forty-epoch promoted pair. Paid compute is forbidden until a runtime and cost estimate receives fresh authorization.
Status	queued

```yaml
budgets:
  local_smoke_samples: 128
  short_epochs: 5
  short_candidate_pairs_max: 3
  final_epochs: 40
  seeds_default: 1
  runpod_total_usd: pending_fresh_authorization
scope:
  collection: spiking-heidelberg-digits
  experiment_id: exp070
  activity_trace: ar073
  tools_snn_edits: narrow_and_backward_compatible_only
  official_test_access: forbidden
  pr: 52
  main_edits: forbidden
stop_when:
  - first_candidate_promoted_and_final_pair_complete
  - all_candidates_killed
  - paid_compute_authority_required
  - protocol_integrity_failure
  - invasive_engine_change_required
```

Paid compute begins only after the local checks pass and the scientist approves an explicit estimate and hard ceiling. At most one pod per cell may be used under that approval, and every pod must be reaped immediately. Do not run the full test suite on the local 4 GB host. Use focused checks and the complete Demolab build.

Record

2026-07-19 06:44–06:49 UTC: diagnosis and registration

The scientist approved the exploratory objective, one-seed policy, locked comparison, intervention order, official-test prohibition, and fresh paid-compute gate in the attached goal. The existing pull request remains the requested programme branch; `main` is not modified. This branch checkpoint therefore serves as the pre-result anchor, a documented workflow deviation from placing a new-night mandate on `main`.

Exp069′s complete local histories show finite but slow learning in both cells, active populations, and no skipped updates. Its COBA input gradient is repeatedly clipped while PING's dampened gradient is much smaller, yet both validation losses continue falling. A development-data-only input audit found that extending the one-second window would recover too few events to explain the accuracy ceiling. Coarsening to 2 ms loses little samechannel event multiplicity and halves the recurrent unroll; 4 ms is already coarse relative to the AMPA decay. Candidate 1 was therefore fixed at 2 ms before any exp070 training result existed. No new runner, smoke, cloud job, official-test access, or spend exists at this checkpoint.

2026-07-19 07:00–07:04 UTC: candidate-1 smoke gate

The 2 ms candidate passed its registered local plumbing smoke. Both cells completed two finite epochs on 128 development-training and 128 validation utterances with the exp069 partition hashes and no skipped or non-finite updates. COBA's final excitatory rate was 16.18 Hz. PING's final excitatory and inhibitory rates were 4.27 Hz and 15 Hz. The trainer used an experiment-specific development-data alias, so the official test cache was neither copied nor opened. Cloud spend remained 0 USD and no pod was created.